

Syed Maoz Arif

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PROFESSIONAL SUMMARY

Electrical Engineering undergraduate (4th semester) specialising in multi-layer PCB design, embedded systems, and IoT. Altium Global Scholarship recipient (2026) and Instructables Sensors Contest Runner-Up (2026). Active freelance hardware & mechanical designer on Fiverr and Upwork, delivering DFM-ready PCBs and 3D assemblies to international clients. Community leader as Membership Head for IET On Campus and Cabinet Member of the Youth Parliament Pakistan. Eager to apply proven hardware and software expertise in a professional engineering role.

EDUCATION

B.Sc. Electrical (Electronics) Engineering

COMSATS University Islamabad, Abbottabad Campus

2024 – 2028 (Expected)

Student ID: FA24-EEE-014

- Currently in 4th Semester. Coursework: Digital Logic Design, Embedded Systems, Circuit Theory, Signals & Systems.
- Altium Global Scholarship for Emerging Engineers (2026) — awarded internationally to outstanding hardware design students.

FSc Pre-Engineering

Peace Group of School & College, Abbottabad

Completed 2024

- Focus areas: Mathematics, Physics, Analytical Reasoning.

TECHNICAL SKILLS

PCB / ECAD	Altium Designer, Altium 365, CircuitStudio, KiCad, EasyEDA — multi-layer routing, schematic capture, DRC, Gerber output
Simulation	Proteus, LTspice, Multisim
Embedded / IoT	ESP32, Arduino Nano, STM32, MPU6050; C/C++ firmware; Firebase, Blynk, MQTT
CAD / 3D	Autodesk Fusion 360, Inventor Professional, SolidWorks, AutoCAD, Revit, Tinkercad — parametric modelling, 3D-print preparation
Programming	C/C++, Python (OOP, automation), MATLAB, HTML5/CSS3/JavaScript, LaTeX / Overleaf
App & Web Dev	Flutter (cross-platform UI), HTML5, CSS3, Vanilla JavaScript, Firebase Cloud Firestore, responsive UI/UX, SEO
AI & ML Tools	Gemini, ChatGPT, Claude, Perplexity, Grok, Google AI Studio, Microsoft Copilot
Design & Docs	Adobe Photoshop, Canva, Inkscape, draw.io, LaTeX / Overleaf
Productivity	Microsoft Office (Word, Excel, PowerPoint, OneNote), Microsoft Teams, Zoom, Git, VS Code

PROJECTS

IncliSense — Precision Digital Level [Runner-Up, Instructables Sensors Contest 2026]

2026

Arduino Nano · MPU6050 · 3D Printing · IoT

- Built a sub-degree-resolution digital inclinometer using MPU6050 IMU with dual-axis tilt compensation, 11-LED bar-graph display, and PC telemetry.
- Delivered custom 3D-printed enclosure, open-source firmware, and full documentation; recognised internationally on Instructables.

Omni-Sense Node — Centralised DAQ & Control System

2025

ESP32 · Altium Designer · Firebase · C++

- Designed a multi-layer PCB integrating 15 peripherals (RFID, gas sensors, servos) on a single ESP32 with real-time Firebase telemetry.
- Engineered dual-rail power distribution, ADC1-mapped analog channels, and ground-plane noise management for reliable mixed-signal operation.

Personal Portfolio Website

2025 – Present

HTML5 · CSS3 · JavaScript · Firebase Firestore

- Developed a responsive portfolio site with Firebase-backed real-time likes/comments, glassmorphism UI, scroll animations, and an AI-assistant simulation mode.
- Deployed on GitHub Pages with full SEO (structured data, Open Graph, sitemap). **Live:** buismaoz346-bit.github.io/my-website

Mixed-Signal RF Transceiver Board

2025

ESP32 · Altium Designer · RF TX/RX · 50Ω Matching

- Laid out an RF transceiver PCB with 50Ω controlled-impedance traces, coplanar waveguide via-fencing, and a 312-via stitched Faraday cage.
- Co-located digital ESP32 and sensitive RF chains with strict copper keepout zones to preserve wireless and RF performance.

Smart Lighting System with Obstacle Detection

2025

Altium Designer · ATMEGA328PB-AU · VL53L0X ToF · USB-C

- Designed a custom PCB with ATMEGA328PB-AU and VL53L0X Time-of-Flight sensor for distance-based automated dual-LED lighting.
- Integrated TP4056 USB-C charging circuit for safe lithium-ion power management.

6S UAV Battery Management System

2025

BQ76930 · XT60 · MOSFETs · Altium Designer

- Designed a 6-series Li-ion BMS schematic using BQ76930 for cell-level monitoring, balancing, and hardware-level safety via dedicated charge/discharge MOSFETs.

Power Regulator and LED Board

2025

Altium Designer · Regulator IC

- Designed a voltage regulation PCB with dual-layer polygon-pour heatsinking, EMI shielding, and low-impedance return paths for clean DC output.

Wi-Fi Router UPS Power Bank

2025

3S Li-ion · BMS · DC-DC Conversion

- Built a custom DC UPS with 3S 12.6 V Li-ion pack, BMS, overcurrent fuse, reverse-polarity protection, and precision-calibrated 12 V output for uninterrupted router power.

EXPERIENCE & LEADERSHIP

Freelance Hardware & Mechanical Designer

2024 – Present

Fiverr & Upwork

- End-to-end PCB design for international clients: schematic capture, multi-layer routing, DFM review, Gerber delivery.
- Parametric 3D product models and manufacturing-ready assemblies using Autodesk Fusion 360 and Inventor.

Membership Head

May 2025 – Present

IET On Campus, Abbottabad

- Leading membership initiatives and organising technical events, including coordinating workshops like Computer Vision & AI in UAVs.
- Directed student recruitment and technical seminars, expanding active membership and industry engagement.

Cabinet Member

2025 – 2027 Session

Youth Parliament Pakistan

- Represent KPK in national policy simulations; drafted resolutions on technology, education, and youth development.

Awardee — Altium Global Scholarship Program

March 2026

- Selected as an emerging engineer for professional development and technical excellence in hardware design.

Runner-Up — Instructables Sensors Contest 2026

April 2026

- Recognised globally for designing and developing the “IncliSense” precision digital level project.

CERTIFICATIONS

- Altium Certified: Designer Essentials — *Altium* (April 5, 2026)
- Altium Education PCB Basic Design Course — *Altium Education* (2025)
- Arduino & IoT — *Cardiff Institute* (2026)
- Generative AI Mastermind — *Outskill* (2025)
- Unleashing Creativity with AI (2026)
- Practical Entrepreneurship (2026)
- 3D Printing Certificate (2026)
- Solar Solution for Pakistan (2026)
- IET On Campus Leadership Recognition — *IET* (2026)
- IET On Campus Community Recognition — *IET* (2026)
- Student Week SP26 Certificate (2026)
- Kangaroo Test Achievement (2026)
- Inovatex Project 2025 (2026)
- Best Project Award — *COMSATS University* (2026)

AWARDS

- **Altium Global Scholarship for Emerging Engineers** — Awarded internationally for outstanding hardware design (March 2026).
- **Runner-Up — Instructables Sensors Contest 2026** — Bronze medal for the “IncliSense” precision digital level project (April 2026).
- **1st Position — Best Circuit Design** — Competitive hardware workshop at COMSATS University × IET (2026).
- **Best Project Award** — Outstanding engineering design and technical execution, COMSATS University (2026).

SOFT SKILLS

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|----------------------------|---------------------------------------|
| • Public Speaking | • Leadership & Project Management |
| • Teamwork & Collaboration | • Critical Thinking & Problem Solving |
| • Communication & Empathy | • Creativity & Organisation |

INTERESTS

Cricket · Trekking · Exploring New Places